

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
# Load packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(trend)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
# Check working directory
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
# Set as default theme
my_theme <- theme_minimal() +
  theme(
    # text
    plot.title = element_text(face = "bold", hjust = 0.5, size = 14),
    axis.title.y = element_text(size = 12, margin = margin(r = 8)),
    axis.text = element_text(size = 10),
    # lines & grids
    axis.ticks = element_line(linewidth = 0.3),
    panel.grid.major = element_line(linewidth = 0.2, linetype = "dashed", color = "grey75"),
    panel.grid.minor = element_blank(),
    # backgrounds
    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = "grey85"),
    legend.key = element_rect(fill = "white", color = NA),
  )
theme_set(my_theme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
EPAair1 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"), stringsAsFactor=
EPAair2 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"), stringsAsFactor=
EPAair3 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"), stringsAsFactor=
EPAair4 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"), stringsAsFactor=
EPAair5 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"), stringsAsFactor=
EPAair6 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"), stringsAsFactor=
EPAair7 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"), stringsAsFactor=
EPAair8 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"), stringsAsFactor=
EPAair9 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"), stringsAsFactor=
EPAair10 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"), stringsAsFactor=
```

```
GaringerOzone <- bind_rows(EPAair1, EPAair2, EPAair3, EPAair4, EPAair5,
                           EPAair6, EPAair7, EPAair8, EPAair9, EPAair10)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3
GaringerOzone <- GaringerOzone %>%
  mutate(Date = as.Date(Date, format = "%m/%d/%Y"))

# 4
GaringerOzone_subset <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
# Create a sequence of all days from 2010-01-01 to 2019-12-31
Days <- as.data.frame(seq.Date(from = as.Date("2010-01-01"),
                               to   = as.Date("2019-12-31"),
                               by   = "day"))

# Rename the column to "Date"
colnames(Days) <- "Date"

# 6
GaringerOzone <- Days %>%
  left_join(GaringerOzone_subset, by = "Date")

summary(GaringerOzone)
```

##	Date	Daily.Max.8.hour.Ozone.Concentration	DAILY_AQI_VALUE
##	Min. :2010-01-01	Min. :0.00200	Min. : 2.00
##	1st Qu.:2012-07-01	1st Qu.:0.03200	1st Qu.: 30.00
##	Median :2014-12-31	Median :0.04100	Median : 38.00
##	Mean :2014-12-31	Mean :0.04163	Mean : 41.57
##	3rd Qu.:2017-07-01	3rd Qu.:0.05100	3rd Qu.: 47.00
##	Max. :2019-12-31	Max. :0.09300	Max. :169.00
##		NA's :63	NA's :63

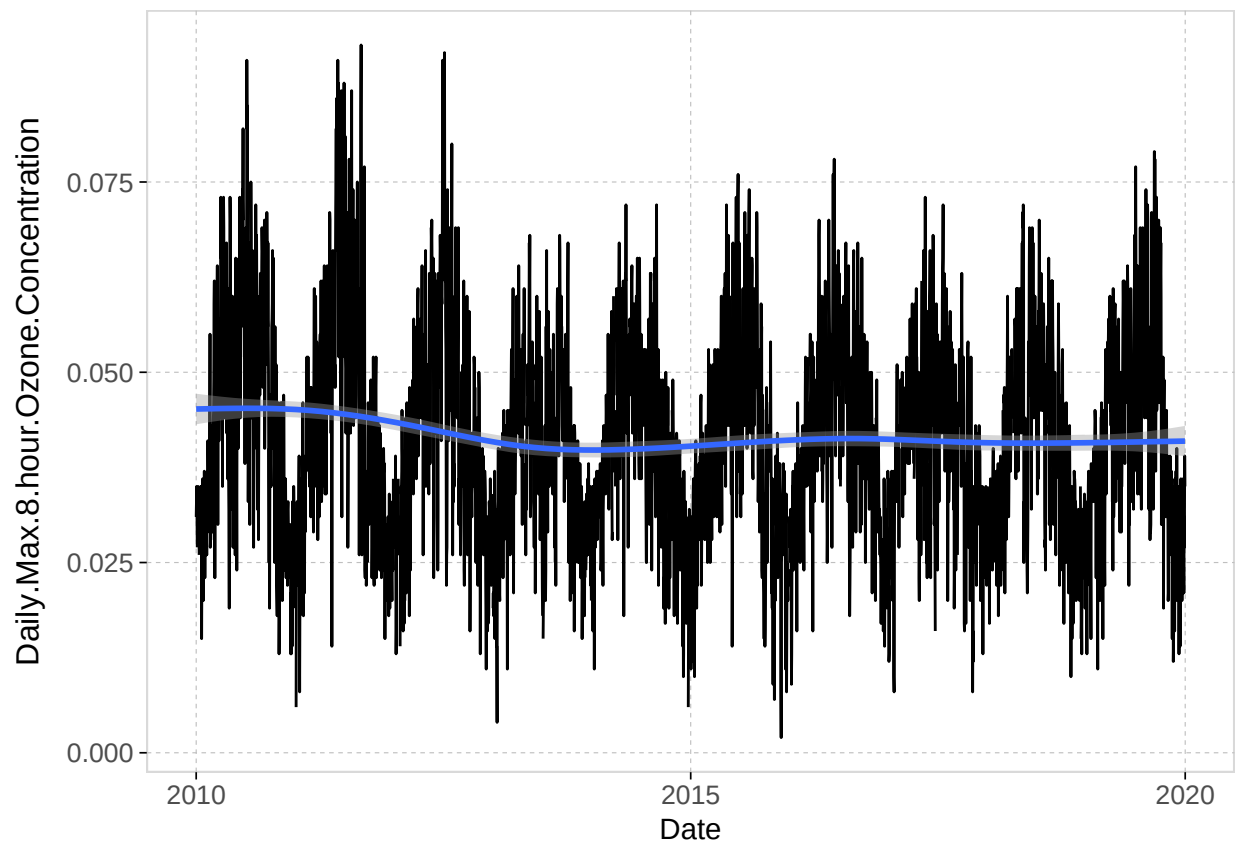
Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
ggplot(GaringerOzone, aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line() +
  geom_smooth()

## 'geom_smooth()' using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: The plot shows clear yearly fluctuations in ozone levels, with higher concentrations during the warmer months. The smoothed line suggests a small decline from 2010 to around 2015, followed by relatively steady levels through 2019. Overall, ozone concentrations appear to have slightly decreased over the decade.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
```

```
head(GaringerOzone)
```

```
##           Date Daily.Max.8.hour.Ozone.Concentration DAILY_AQI_VALUE
## 1 2010-01-01                0.031                29
## 2 2010-01-02                0.033                31
## 3 2010-01-03                0.035                32
## 4 2010-01-04                0.031                29
## 5 2010-01-05                0.027                25
## 6 2010-01-06                NA                 NA
```

```
summary(GaringerOzone$Daily.Max.8.hour.Ozone.Concentration)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.      NA's
## 0.00200 0.03200 0.04100 0.04163 0.05100 0.09300        63
```

```
GaringerOzone_clean <-
```

```
  GaringerOzone %>%
```

```
  mutate(Daily.Max.8.hour.Ozone.Concentration_clean = zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration,
```

```
summary(GaringerOzone_clean$Daily.Max.8.hour.Ozone.Concentration_clean)
```

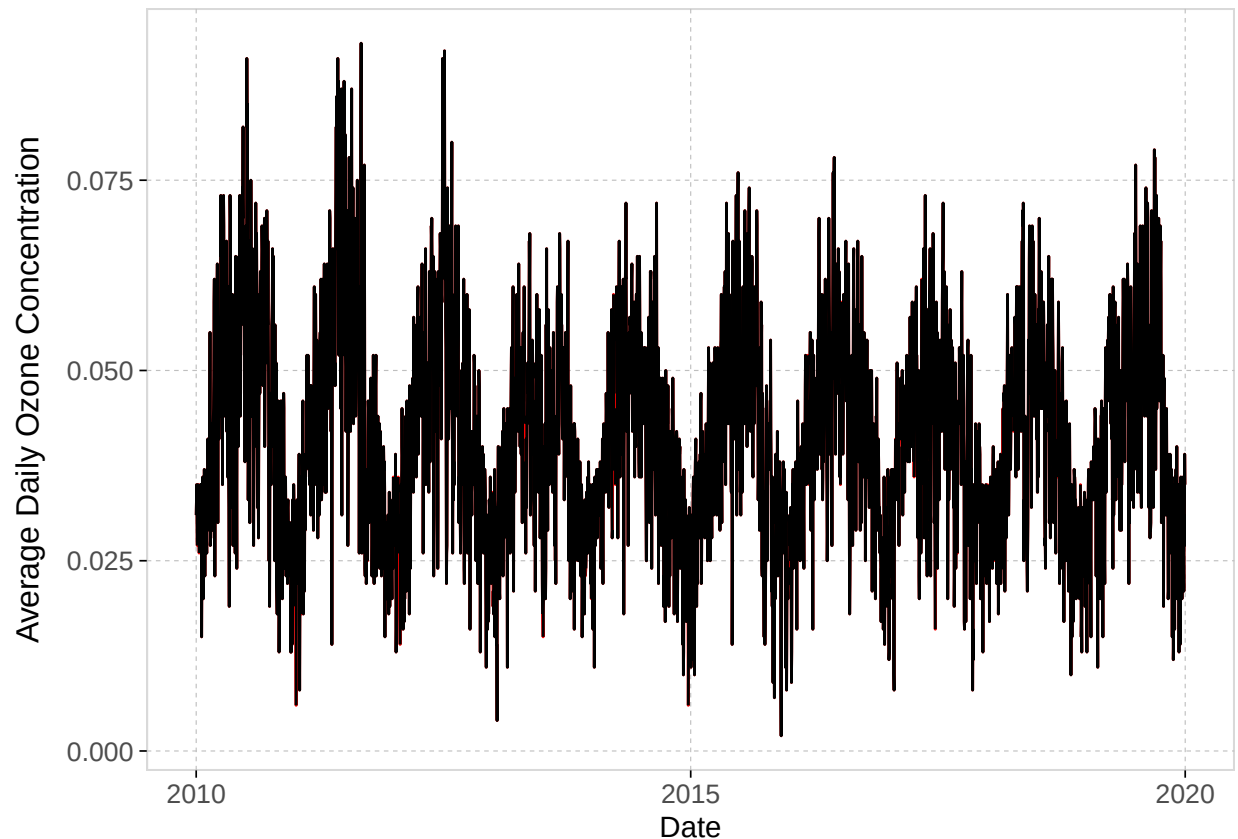
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00200 0.03200 0.04100 0.04151 0.05100 0.09300
```

```
ggplot(GaringerOzone_clean) +
```

```
  geom_line(aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration_clean), color = "red") +
```

```
  geom_line(aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration), color = "black") +
```

```
  ylab("Average Daily Ozone Concentration")
```



Answer: We use linear interpolation (`na.approx`) because it provides a simple, physically reasonable estimate for missing daily ozone values without distorting real patterns. Piecewise constant would ignore gradual changes, and spline interpolation could over-smooth or introduce artificial variability.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new `Date` column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(
    Year = year(Date),
    Month = month(Date)
  ) %>%
  group_by(Year, Month) %>%
  summarise(
    MeanOzone = mean(Daily.Max.8.hour.Ozone.Concentration, na.rm = TRUE)
  ) %>%
  ungroup()
```

```
## 'summarise()' has grouped output by 'Year'. You can override using the
## '.groups' argument.
```

```
GaringerOzone.monthly <- GaringerOzone.monthly %>%
  mutate(Date = as.Date(paste(Year, Month, "01", sep = "-")))
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10
# Daily time series
f_daily <- c(year(min(GaringerOzone_clean$Date)),
             yday(min(GaringerOzone_clean$Date)))

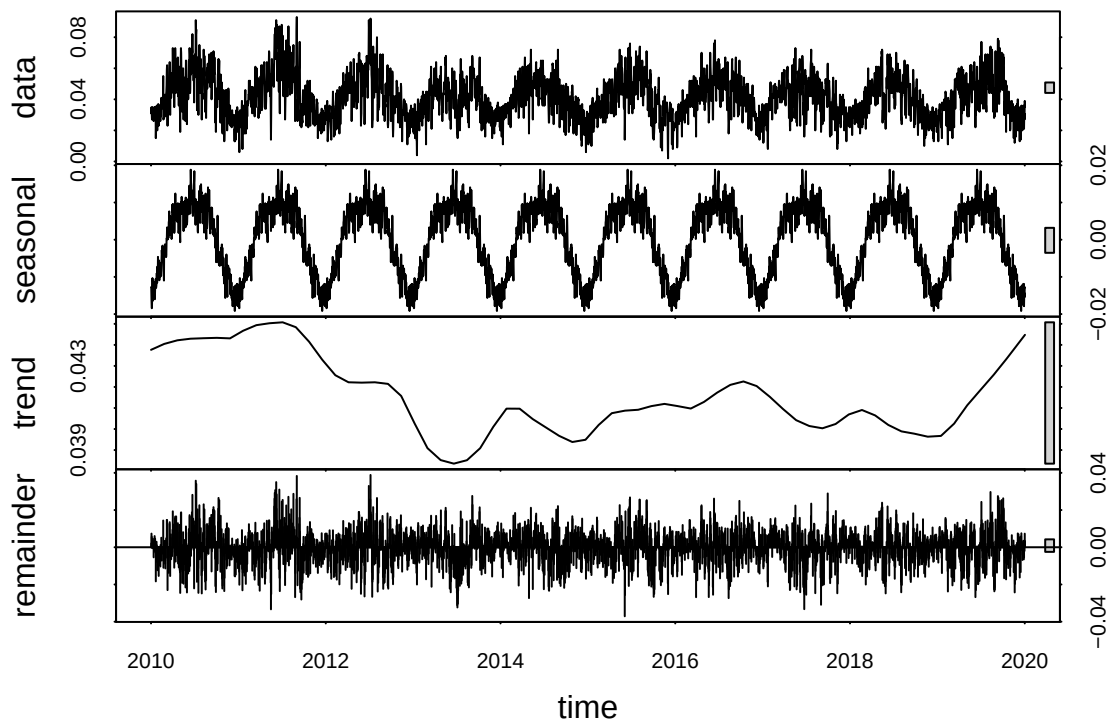
GaringerOzone.daily.ts <- ts(
  GaringerOzone_clean$Daily.Max.8.hour.Ozone.Concentration.clean,
  start = f_daily, frequency = 365
)

# Monthly time series
f_month <- c(min(GaringerOzone.monthly$Year),
             min(GaringerOzone.monthly$Month))

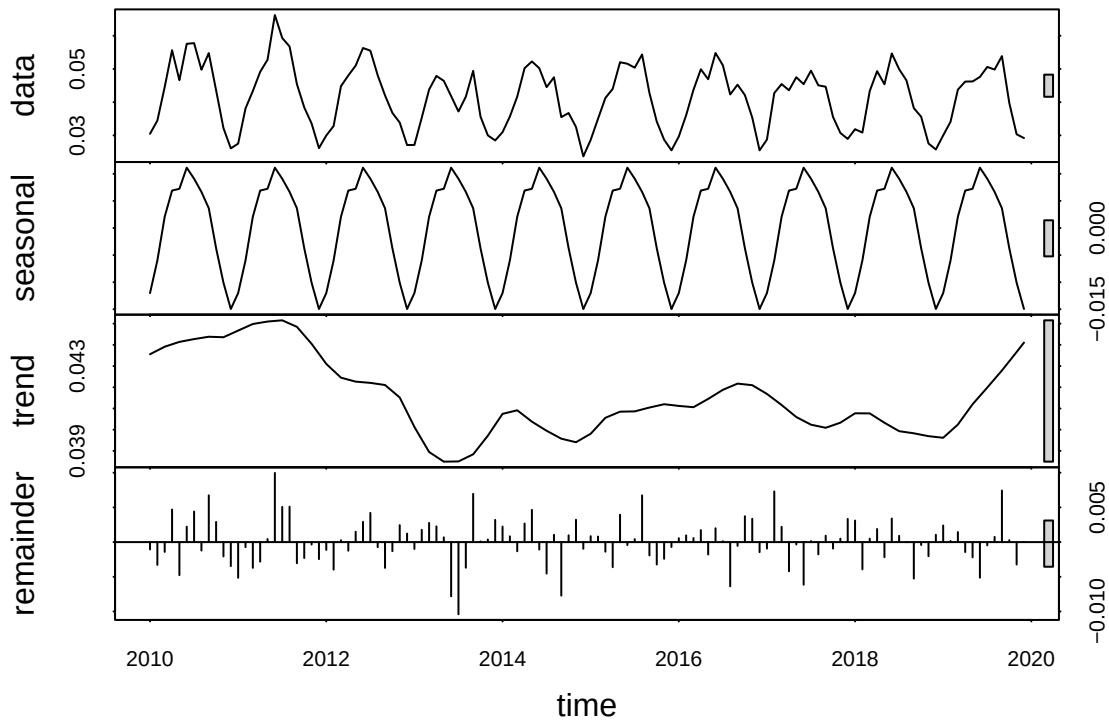
GaringerOzone.monthly.ts <- ts(
  GaringerOzone.monthly$MeanOzone,
  start = f_month, frequency = 12
)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
# Daily time series plot
GaringerOzone_daily_Decomposed <- stl(GaringerOzone.daily.ts, s.window = "periodic")
plot(GaringerOzone_daily_Decomposed)
```



```
# Monthly time series plot
GaringerOzone_monthly_Decomposed <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(GaringerOzone_monthly_Decomposed)
```

12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

#12

Run SMK test

```
GaringerOzone_trend1 <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)
```

Inspect results

```
GaringerOzone_trend1
```

```
## tau = -0.163, 2-sided pvalue =0.022986
```

```
summary(GaringerOzone_trend1)
```

```
## Score = -88 , Var(Score) = 1498
```

```
## denominator = 538.9944
```

```
## tau = -0.163, 2-sided pvalue =0.022986
```

```
GaringerOzone_trend2 <- trend::smk.test(GaringerOzone.monthly.ts)
```

```
GaringerOzone_trend2
```

```
##
```

```
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## z = -2.2478, p-value = 0.02459
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##      S varS
## -88 1498
```

```
summary(GaringerOzone_trend2)
```

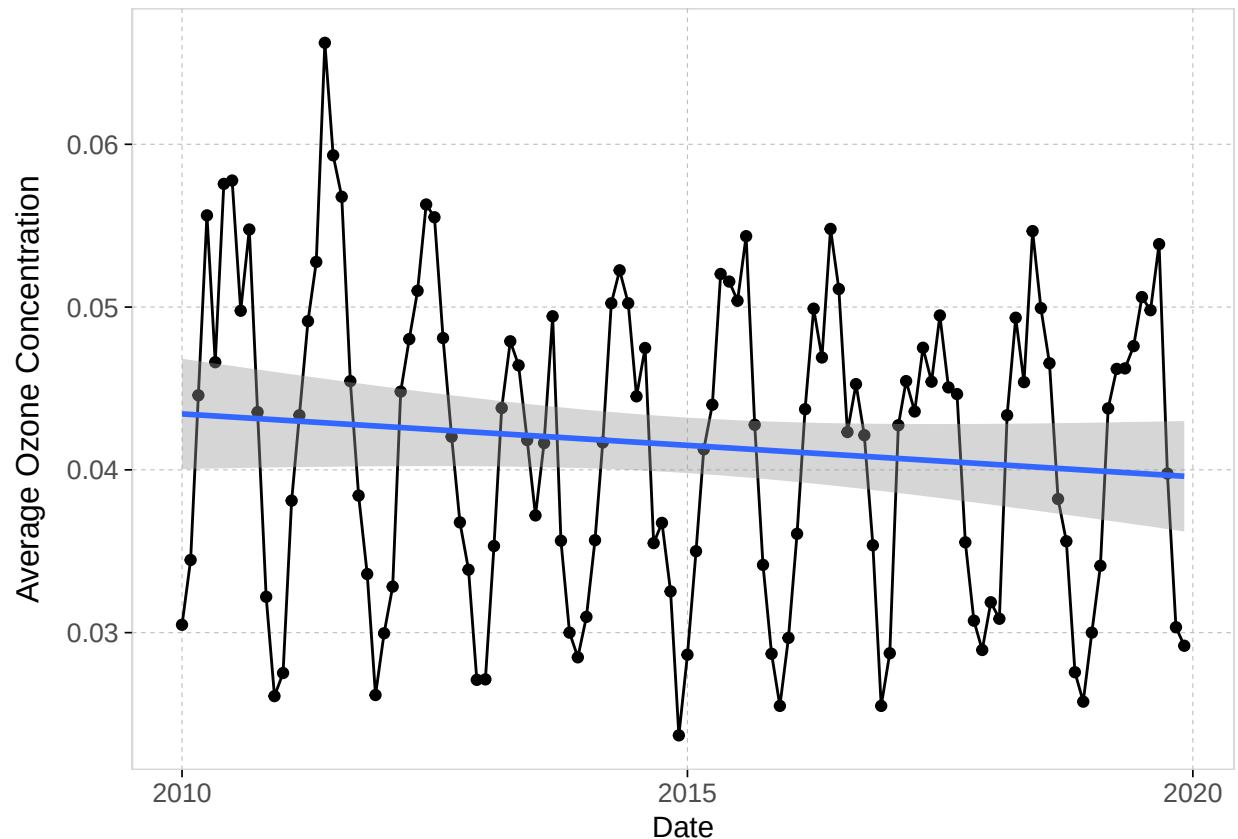
```
##
## Seasonal Mann-Kendall trend test (Hirsch-Slack test)
##
## data: GaringerOzone.monthly.ts
## alternative hypothesis: two.sided
##
## Statistics for individual seasons
##
## H0
##
##      S varS      tau      z Pr(>|z|)
## Season 1:  S = 0      9 125  0.200  0.716  0.47427
## Season 2:  S = 0     -1 125 -0.022  0.000  1.00000
## Season 3:  S = 0     -4 124 -0.090 -0.269  0.78762
## Season 4:  S = 0    -17 125 -0.378 -1.431  0.15241
## Season 5:  S = 0    -17 125 -0.378 -1.431  0.15241
## Season 6:  S = 0    -17 125 -0.378 -1.431  0.15241
## Season 7:  S = 0    -11 125 -0.244 -0.894  0.37109
## Season 8:  S = 0     -7 125 -0.156 -0.537  0.59151
## Season 9:  S = 0     -7 125 -0.156 -0.537  0.59151
## Season 10: S = 0    -13 125 -0.289 -1.073  0.28313
## Season 11: S = 0    -13 125 -0.289 -1.073  0.28313
## Season 12: S = 0     10 124  0.225  0.808  0.41896
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer: The Seasonal Mann Kendall test is most appropriate because ozone levels vary with the seasons. This method accounts for those repeating yearly patterns while testing for an overall trend. The results ($\tau = -0.163$, $p = 0.023$) show a small but significant decrease in monthly ozone concentrations from 2010 to 2019.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
# Visualization
GaringerOzone_trend_plot <-
  ggplot(GaringerOzone.monthly, aes(x = Date, y = MeanOzone)) +
  geom_point() +
  geom_line() +
  ylab("Average Ozone Concentration") +
  geom_smooth(method = lm)
print(GaringerOzone_trend_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: Ozone concentrations at this station show regular seasonal fluctuations, with higher levels during warmer months and lower levels in cooler months. Over the 2010s, the overall trend indicates a small but steady decline in ozone concentrations. The Seasonal Mann Kendall test confirms this downward trend as statistically significant ($\tau = -0.163$, $p = 0.023$).

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
# We can extract the components and turn them into data frames
GraingerOzone_Components <- as.data.frame(GaringerOzone_monthly_Decomposed$time.series[,1:3])

GraingerOzone_Components <- mutate(GraingerOzone_Components,
  Observed = GaringerOzone.monthly$MeanOzone,
  Date = GaringerOzone.monthly$Date)
```

```

GraingerOzone_Components <- GraingerOzone_Components %>%
  mutate(Deseasonal = Observed - seasonal)

#16
GraingerOzone_monthly_deseasonal_ts <- ts(GraingerOzone_Components$Deseasonal,
                                           start = start(GraingerOzone.monthly.ts),
                                           frequency = frequency(GraingerOzone.monthly.ts))

GraingerOzone_Components_trend <- Kendall::MannKendall(GraingerOzone_monthly_deseasonal_ts)

GraingerOzone_Components_trend

## tau = -0.179, 2-sided pvalue =0.0037728

summary(GraingerOzone_Components_trend)

## Score = -1278 , Var(Score) = 194364.7
## denominator = 7139
## tau = -0.179, 2-sided pvalue =0.0037728

```

Answer: Both tests show a significant decline in ozone concentrations over time. The Seasonal Mann Kendall test with tau equal to negative 0.163 and p value 0.023 indicates a negative trend after accounting for recurring seasonal patterns, while the Mann Kendall test on the deseasonalized series with tau equal to negative 0.179 and p value 0.0038 shows an even stronger decrease. This suggests that once seasonal effects are removed, the long term downward trend in ozone levels at the Garinger station becomes more pronounced.